

Ahmed Mohamed Amin

AI Engineer | RAG, NLP and Applied LLMs

✉ ahmed.m.amin2003@gmail.com

☎ +20 128 802 4972

📍 Giza, Egypt

🐙 [GitHub](#)

🌐 [LinkedIn](#)

🌐 [Portfolio](#)

PROFILE

AI Engineer building production retrieval-augmented generation systems, and an M.Sc. student in AI for Science at the African Institute for Mathematical Sciences, fully funded by a Google DeepMind Scholarship. I focus on making large language models cheap enough to run and grounded enough to trust: resource-efficient fine-tuning, hybrid retrieval, and evaluation that shows whether an answer came from its sources. Published at IEEE CSCloud 2025 on modular LoRA fine-tuning. Particular interest in machine learning, NLP and LLM applications, and in Arabic and other low-resource languages.

EDUCATION

African Institute for Mathematical Sciences (AIMS)

Sept 2026 – July 2027 | Cape Town, South Africa

M.Sc. Mathematical Sciences in AI for Science, registered at the University of Cape Town

Google DeepMind Scholarship (full funding): tuition, accommodation, meals, medical cover, flights, stipend and research compute

University of Science and Technology at Zewail City

Sept 2021 – June 2026 | Giza, Egypt

B.Sc. Communication & Information Engineering (ABET accredited)

GPA: 3.64/4.00, Cum Laude. Provost's Honors Roll, 7 semesters

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Big Data Analytics, Statistical Inference, Algorithms & Data Structures, Linear Algebra

PROFESSIONAL EXPERIENCE

AI Engineer

Dec 2025 – Present

Takhaial, Giza, Egypt

- Shipped **four client-facing AI assistants** to paying customers, owning each from API design through deployment (detailed below).
- Cut inference cost per message by roughly **5×**, from about 5 EGP to 1 EGP, by redesigning the embedding and retrieval layer.
- Raised answer quality using hybrid BM25 and vector retrieval with RRF fusion and MMR reranking, plus query routing that selects the right knowledge source for each question.
- Designed an **asynchronous architecture** on Celery running speech-to-text, text-to-speech and retrieval in parallel, so one API serves both voice and text conversations.
- Deployed containerised microservices (FastAPI, ChromaDB, MongoDB, Redis, MinIO) on AWS EC2 with Docker Swarm and zero-downtime releases, monitored through a Promtail/Loki tracing dashboard.
- Lead a two-person engineering team:** assign work, review delivery, and report progress monthly.

PUBLICATION

FinVerse Agent: Resource-Efficient Financial LLMs with Modular LoRA Fine-Tuning and Task Routing

Aug 2025

Proc. IEEE CSCloud 2025, 12th Int. Conference on Cyber Security and Cloud Computing

- L. Rana, **A. M. Amin**, H. Ameer, and S. Latif. Available on [IEEE Xplore](#); adapter code on [GitHub](#).
- My contribution:** second author, and the Zewail City side of a collaboration with NUST Islamabad. Fine-tuned 15 task-specific LoRA adapters on a 4-bit quantised LLaMA-3.1-8B (**10×** cheaper than full fine-tuning) and built the Planner–Executor–Verifier loop that picks an adapter per task and validates its output.
- Results:** **93.07%** on CFA Level I (GPT-4o: 88.10%) and **97.24 F1** on financial NER (BloombergGPT: 60.82), using 75% less memory. **Won 1st Place, FinAI Contest 2025 (Task 2).**

GRADUATION PROJECT

Rafeeq: Modular Intelligent System for Self-Navigation and Health Monitoring

Sept 2025 – 2026

Lead AI Architect. Arabic NLP, Robotics, Edge AI

- Designed a kit adding autonomous navigation, Arabic voice control and health monitoring to any manual wheelchair, at **60–70% below** the cost of an imported smart wheelchair.
- Built an offline Egyptian Arabic speech pipeline (faster-whisper INT8) on a Raspberry Pi 5, reaching **87% command accuracy at 0.9s latency** with no internet connection.
- Added indoor navigation using LiDAR and depth-camera SLAM (ROS 2 Nav2), with a three-tier controller converting commands into motor movement in under 20 ms.
- Awarded a **\$1,600 grant** from Egypt's Ministry of Higher Education, selected nationally across all Egyptian universities.

SELECTED PROJECTS

Production RAG assistants at Takhaial. Full write-ups on the [portfolio](#).

- Dr. Sherine, Arabic advisory voice agent (write-up).** Built retrieval over a fully diacritized Classical Arabic corpus for users who type undiacritized colloquial Egyptian. Quotes source text verbatim before explaining it in dialect, and says plainly when the corpus cannot answer. Indexed **23,512 chunks across 9 books** and scored answers against a **100-question evaluation set**.

- **The Spine, multilingual property assistant** ([write-up](#), [multi-channel extension](#)). Shipped a buyer-facing assistant that answers from an approved knowledge base and attaches the right floor plans and renders, then extended it to WhatsApp, Messenger and Instagram. Owned the retrieval and image layer, the multilingual output, and the rate limiting that keeps a business account from being flagged. Tested across Arabic, English, Spanish and French over a **702-image** catalogue on the web assistant and **845 indexed images** across 43 buildings on the messaging channels, with a **66-case evaluation harness**.
- **Trading and market assistant** ([write-up](#)). Built an English and **Gulf-Arabic** assistant for an online brokerage covering live market data, product questions, lead capture, and voice over both web and phone. **~11,500 lines** across 19 services on a 9-container AWS stack.

Machine learning, forecasting and vision

- **Multi-camera threat detection** ([write-up](#), [code](#)). Sole engineer on a real-time system watching several camera feeds for weapons, falls and violence, with automatic evidence capture and escalation. Trained the models and built the architecture, threading model, alerting and dashboard, reaching **0.844 mAP@0.50** on weapon detection and 81% on violence, at 698 ms mean CPU latency.
- **Time-series load forecasting and battery dispatch** ([write-up](#), [code](#)). Trained LightGBM day-ahead load forecasters at **96 fifteen-minute steps** per day, including generalisation to an unseen site, then fed the forecasts to a rolling-horizon MPC solving a linear program at every step to schedule a 16 kWh battery against a 9 kWp array on the real Italian time-of-use tariff. Landed within **€0.33** of a perfect-information oracle and **€10.51** ahead of the site controller.
- **Deep spatio-temporal networks for CTR prediction** ([code](#)). Reimplemented the D-STN architecture from the KDD 2019 paper by Ouyang et al. in TensorFlow, modelling auxiliary spatial and temporal context alongside the target instance, and reproduced the paper's reported comparisons.
- **Credit approval prediction** ([write-up](#)). Built an end-to-end classifier for loan repayment risk: cleaned and encoded the credit records, engineered a weighted credit-history score, applied SMOTE to correct the class imbalance, and pruned zero-importance features using XGBoost importances. Compared six models and tuned the best by grid-searched cross-validation, reaching **95.6% accuracy** with XGBoost against a 58.5% logistic-regression baseline. Coursework for Machine Learning, grade A-.

TECHNICAL SKILLS

- **Programming:** Python, C++, SQL
- **AI & NLP:** RAG, Hybrid Retrieval (BM25 + Vector), RRF, MMR Reranking, PyTorch, Hugging Face Transformers, LangChain, LoRA/QLoRA, PEFT, 4-bit Quantisation, vLLM, Whisper/faster-whisper, Computer Vision
- **Data & Retrieval:** PostgreSQL/pgvector, ChromaDB, FAISS, SQLite FTS5, MongoDB, Redis
- **Infrastructure:** AWS EC2, Docker, Docker Swarm, Celery, Promtail/Loki, Linux, Git, CI/CD
- **Software Engineering:** FastAPI, REST API design, microservices, system design, ROS 2

SCHOLARSHIPS & AWARDS

- **Google DeepMind Scholarship**, full funding for the AI for Science M.Sc. **2026**
- **1st Place, FinAI Contest 2025** (Task 2), out of the FinVerse Agent work above **2025**
- **Ministry of Higher Education Grant**, Innovators Support Fund, for Rafeeq **2026**
- **Full Excellence Scholarship**, five years of full tuition at Zewail City **2021–2026**

LEADERSHIP & TEACHING

Elected Student Representative **2021 – 2025**

University of Science and Technology at Zewail City, Giza, Egypt

- Represented all four CIE cohorts for four consecutive years, working with faculty on academic matters affecting **200+ students**.

Junior Teaching Assistant, Math 101 (Calculus I) **Oct 2022 – Jan 2023**

University of Science and Technology at Zewail City, Giza, Egypt

- Led tutorials and one-on-one mentoring for **50+ first-year students** in calculus and linear algebra.

Egypt Scholars, Basic Lab 12 **Feb 2026 – May 2026**

Competitively selected participant

- Selected for a three-month programme on research methods, scientific writing and critical thinking.

LANGUAGES

Arabic: Native (MSA and Egyptian dialect)

English: Professional proficiency (IELTS Academic 6.5, CEFR B2)